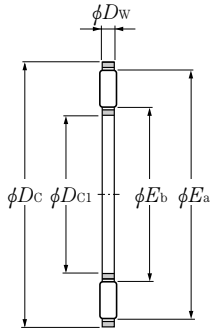
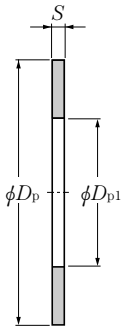
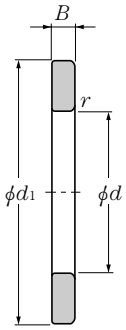
Type AXK11
Type AS11
Type WS811
Type GS811



Type AXK (Thrust needle roller and cage assy)



AS bearing ring (washer)

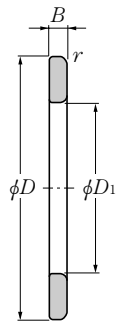


WS bearing ring (Inner ring)

D_{c1} 150～160mm

Boundary dimensions mm												Basic load ratings N		kgf	
												dynamic	static	dynamic	static
D_{c1}	D_c	D_w	D_p	D_{p1}	S 2)	d	d_1	D	D_1	B	$r_{s\,min}$ 1)	C_a	C_{oa}	C_a	C_{oa}
E11	c12	$^{0}_{-0.01}$	e13	E12	0.05		$^{-0.2}_{-0.5}$		$^{+0.5}_{+0.2}$						
150	190	5	190	150	1	150	188	190	152	$9.5\,^{0}_{-0.090}$	1	149 000	1 020 000	15 200	104 000
160	200	5	200	160	1	160	198	200	162	$9.5\,^{0}_{-0.090}$	1	154 000	1 070 000	15 700	110 000

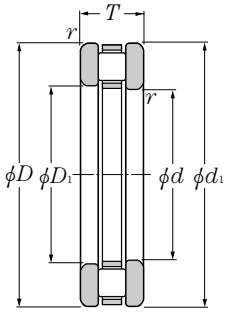
Note 1) Allowable minimum chamfer dimension r.
2) Subject to measured thrust load of 20kg or more.



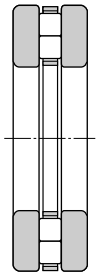
GS bearing ring
(Outer ring)

Limiting speeds r/min		Bearing numbers				Reference dimensions mm		Mass kg (approx.)		
grease	oil	thrust needle roller and cage assembly	washer	inner ring	outer ring	E_b	E_a	AXK11	AS11	WS811 GS811
350	1 400	**AXK1130**	**AS1130**	**WS81130**	**GS81130**	156.0	184.2	0.232	0.084	0.796
330	1 300	**AXK1132**	**AS1132**	**WS81132**	**GS81132**	166.0	194.2	0.246	0.089	0.842

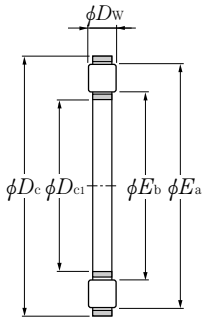
Type 811
Type 812
Type 893



Type 811
Type 812
(Bearing)



Type 893
(Bearing)



Type K811, type K812
(Thrust needle roller and cage assy)

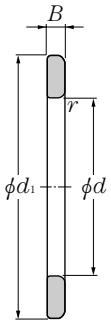
d 10～60mm

Boundary dimensions mm										Basic load ratings N		Basic load ratings kgf		Limiting speeds r/min	
d	D	d_1 $^{-0.2}_{-0.5}$	D_1 $^{+0.5}_{+0.2}$	T	D_{c1} [2] E11	D_c a13	D_w $^{0}_{-0.010}$	B h11	$r_{s\,min}$ [1]	dynamic C_a	static C_{oa}	dynamic C_a	static C_{oa}	grease	oil
10	24	24	10	9	10	24	3.5	2.75	0.3	10 800	21 500	1 110	2 190	3 400	13 000
12	26	26	12	9	12	26	3.5	2.75	0.3	11 500	23 900	1 170	2 430	3 000	12 000
15	28	28	16	9	15	28	3.5	2.75	0.3	12 900	28 600	1 310	2 920	2 800	11 000
17	30	30	18	9	17	30	3.5	2.75	0.3	13 400	31 000	1 370	3 150	2 500	10 000
20	35	35	21	10	20	35	4.5	2.75	0.3	20 200	46 500	2 060	4 700	2 100	8 500
25	42	42	26	11	25	42	5	3	0.6	27 300	68 000	2 780	6 900	1 800	7 000
	47	47	32	11	30	47	5	3	0.6	27 800	72 500	2 840	7 400	1 500	6 000
30	52	52	32	16	30	52	7.5	4.25	0.6	53 000	129 000	5 450	13 100	1 500	6 000
	60	60	32	18	30	60	5.5	6.25	1	54 000	166 000	5 500	16 900	1 300	5 000
	52	52	37	12	35	52	5	3.5	0.6	31 000	87 000	3 150	8 900	1 400	5 500
35	62	62	37	18	35	62	7.5	5.25	1	54 500	139 000	5 550	14 200	1 200	4 900
	68	68	37	20	35	68	6	7	1	66 000	214 000	6 750	21 800	1 200	4 600
	60	60	42	13	40	60	6	3.5	0.6	43 000	121 000	4 350	12 400	1 200	4 800
40	68	68	42	19	40	68	9	5	1	74 500	190 000	7 600	19 400	1 100	4 400
	78	78	42	22	40	78	7	7.5	1	85 000	277 000	8 700	28 300	1 000	4 000
	65	65	47	14	45	65	6	4	0.6	45 500	135 000	4 650	13 800	1 100	4 400
45	73	73	47	20	45	73	9	5.5	1	82 000	222 000	8 350	22 600	1 000	4 100
	85	85	47	24	45	85	7.5	8.25	1	102 000	345 000	10 400	35 000	900	3 600
	70	70	52	14	50	70	6	4	0.6	48 000	150 000	4 900	15 300	1 000	4 000
50	78	78	52	22	50	78	9	6.5	1	85 000	238 000	8 650	24 200	950	3 800
	95	95	52	27	50	95	8	9.5	1.1	125 000	445 000	12 700	45 000	800	3 200
	78	78	57	16	55	78	6	5	0.6	62 500	215 000	6 350	21 900	900	3 600
55	90	90	57	25	55	90	11	7	1	121 000	340 000	12 300	34 500	830	3 300
	105	105	57	30	55	105	9	10.5	1.1	158 000	570 000	16 100	58 000	730	2 900
	85	85	62	17	60	85	7.5	4.75	1	69 000	215 000	7 000	21 900	830	3 300
60	95	95	62	26	60	95	11	7.5	1	125 000	365 000	12 800	37 000	780	3 100
	110	110	62	30	60	110	9	10.5	1.1	162 000	600 000	16 500	61 500	680	2 700

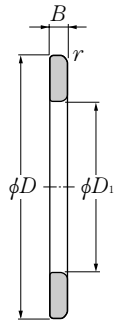
Note 1) Allowable minimum chamfer dimension r .
2) The dimensional tolerance for a bearing with a T2 suffix is E12.



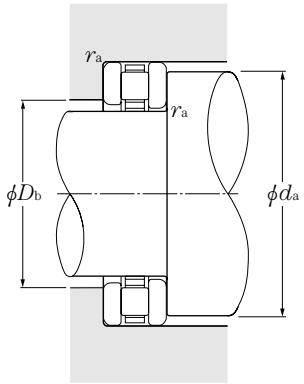
Type K893
(Thrust needle roller and cage assy)



WS bearing ring
(Inner ring)

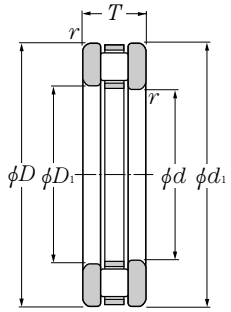


GS bearing ring
(Outer ring)

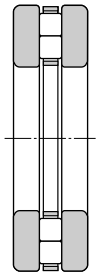


Bearing numbers				Reference dimensions mm		Abutment dimensions mm			Mass kg		
bearing	thrust cylindrical roller and cage assembly	inner ring	outer ring	E_b	E_a	d_a min	D_b max	r_{as} max	811 812 893	K811 K812 K893	WS GS
81100T2	**K81100T2**	**WS81100**	**GS81100**	13.5	21.3	21	14	0.3	0.0195	0.0035	0.008
81101T2	**K81101T2**	**WS81101**	**GS81101**	15.5	23.3	23	16	0.3	0.022	0.004	0.009
81102T2	**K81102T2**	**WS81102**	**GS81102**	17.2	25.0	25	18	0.3	0.026	0.006	0.010
81103T2	**K81103T2**	**WS81103**	**GS81103**	19.2	27.0	27	20	0.3	0.030	0.008	0.011
81104T2	**K81104T2**	**WS81104**	**GS81104**	22.4	32.3	32	23	0.3	0.040	0.012	0.014
81105T2	**K81105T2**	**WS81105**	**GS81105**	27.6	38.7	39	28	0.6	0.060	0.018	0.021
81106T2	**K81106T2**	**WS81106**	**GS81106**	33.1	43.9	44	33	0.6	0.070	0.020	0.025
81206T2	**K81206T2**	**WS81206**	**GS81206**	32.8	49.0	48	33	0.6	0.140	0.050	0.045
89306	**K89306**	**WS89306**	**GS89306**	34.0	56.4	56	34	1	0.250	0.046	0.100
81107T2	**K81107T2**	**WS81107**	**GS81107**	38.0	48.9	49	38	0.6	0.090	0.024	0.033
81207T2	**K81207T2**	**WS81207**	**GS81207**	39.8	56.0	56	41	1	0.235	0.065	0.085
89307	**K89307**	**WS89307**	**GS89307**	40.0	64.4	64	40	1	0.360	0.064	0.150
81108T2	**K81108T2**	**WS81108**	**GS81108**	43.2	56.4	56	44	0.6	0.135	0.035	0.044
81208T2	**K81208T2**	**WS81208**	**GS81208**	43.7	62.9	63	44	1	0.265	0.085	0.090
89308	**K89308**	**WS89308**	**GS89308**	46.0	74.4	74	46	1	0.520	0.100	0.210
81109T2	**K81109T2**	**WS81109**	**GS81109**	48.4	61.6	61	49	0.6	0.150	0.040	0.055
81209T2	**K81209T2**	**WS81209**	**GS81209**	48.8	68.0	68	49	1	0.310	0.100	0.105
89309	**K89309**	**WS89309**	**GS89309**	50.9	81.3	81	51	1	0.670	0.140	0.270
81110T2	**K81110T2**	**WS81110**	**GS81110**	53.2	66.4	66	54	0.6	0.165	0.045	0.060
81210T2	**K81210T2**	**WS81210**	**GS81210**	53.7	73.1	73	54	1	0.385	0.105	0.140
89310	**K89310**	**WS89310**	**GS89310**	58.0	90.4	90	58	1	0.940	0.180	0.380
81111T2	**K81111T2**	**WS81111**	**GS81111**	57.8	75.2	75	58	0.6	0.250	0.060	0.095
81211T2	**K81211T2**	**WS81211**	**GS81211**	60.1	83.4	83	61	1	0.610	0.190	0.210
89311	**K89311**	**WS89311**	**GS89311**	63.9	100.3	100	64	1	1.270	0.240	0.520
81112T2	**K81112T2**	**WS81112**	**GS81112**	63.7	80.1	80	65	1	0.285	0.083	0.101
81212T2	**K81212T2**	**WS81212**	**GS81212**	64.9	88.4	88	66	1	0.660	0.200	0.230
89312	**K89312**	**WS89312**	**GS89312**	68.9	105.3	105	69	1	1.350	0.250	0.550

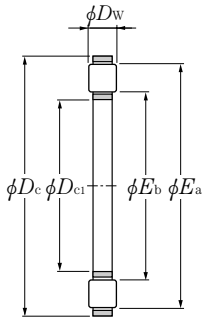
Type 811
Type 812
Type 893



Type 811
Type 812
(Bearing)



Type 893
(Bearing)



Type K811, type K812
(Thrust needle roller and cage assy)

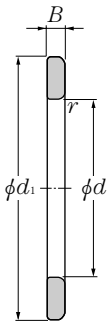
d 65～130mm

Boundary dimensions mm										Basic load ratings dynamic N	Basic load ratings static N	Basic load ratings dynamic kgf	Basic load ratings static kgf	Limiting speeds r/min	
d	D	d_1 $^{-0.2}_{-0.5}$	D_1 $^{+0.5}_{+0.2}$	T	D_{c1} E11	D_c a13	D_w $^{0}_{-0.010}$	B h11	$r_{s\,min}$ [1]	C_a	C_{oa}	C_a	C_{oa}	grease	oil
	90	90	67	18	65	90	7.5	5.25	1	73 000	236 000	7 400	24 100	780	3 100
65	100	100	67	27	65	100	11	8	1	130 000	385 000	13 200	39 500	730	2 900
	115	115	67	30	65	115	9	10.5	1.1	166 000	635 000	17 000	64 500	650	2 600
	95	95	72	18	70	95	7.5	5.25	1	76 500	257 000	7 800	26 200	730	2 900
70	105	105	72	27	70	105	11	8	1	134 000	410 000	13 700	42 000	680	2 700
	125	125	72	34	70	125	10	12	1.1	205 000	790 000	20 900	81 000	600	2 400
	100	100	77	19	75	100	7.5	5.75	1	78 000	268 000	7 950	27 300	680	2 700
75	110	110	77	27	75	110	11	8	1	138 000	435 000	14 100	44 500	650	2 600
	135	135	77	36	75	135	11	12.5	1.5	239 000	920 000	24 400	94 000	550	2 200
	105	105	82	19	80	105	7.5	5.75	1	79 500	279 000	8 100	28 400	650	2 600
80	115	115	82	28	80	115	11	8.5	1	142 000	460 000	14 500	47 000	630	2 500
	140	140	82	36	80	140	11	12.5	1.5	246 000	970 000	25 100	98 500	530	2 100
	110	110	87	19	85	110	7.5	5.75	1	83 000	300 000	8 450	30 500	630	2 500
85	125	125	88	31	85	125	12	9.5	1	169 000	550 000	17 200	56 500	580	2 300
	150	150	88	39	85	150	12	13.5	1.5	281 000	1 100 000	28 600	113 000	500	2 000
	120	120	92	22	90	120	9	6.5	1	112 000	395 000	11 400	40 500	580	2 300
90	135	135	93	35	90	135	14	10.5	1.1	213 000	680 000	21 700	69 500	530	2 100
	155	155	93	39	90	155	12	13.5	1.5	289 000	1 160 000	29 500	118 000	480	1 900
	135	135	102	25	100	135	11	7	1	158 000	555 000	16 100	57 000	500	2 000
100	150	150	103	38	100	150	15	11.5	1.1	243 000	795 000	24 800	81 000	480	1 900
	170	170	103	42	100	170	13	14.5	1.5	335 000	1 370 000	34 500	140 000	430	1 700
	145	145	112	25	110	145	11	7	1	165 000	605 000	16 800	61 500	480	1 900
110	160	160	113	38	110	160	15	11.5	1.1	259 000	885 000	26 400	90 000	450	1 800
	190	190	113	48	110	190	15	16.5	2	430 000	1 770 000	44 000	180 000	400	1 600
	155	155	122	25	120	155	11	7	1	172 000	655 000	17 500	66 500	450	1 800
120	170	170	123	39	120	170	15	12	1.1	264 000	930 000	26 900	94 500	430	1 700
	170	170	132	30	130	170	12	9	1	197 000	755 000	20 100	77 000	400	1 600
130	190	187	133	45	130	190	19	13	1.5	360 000	1 210 000	36 500	123 000	380	1 500

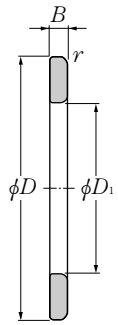
Note 1) Allowable minimum chamfer dimension r .
2) The dimensional tolerance for a bearing with a T2 suffix is E12.



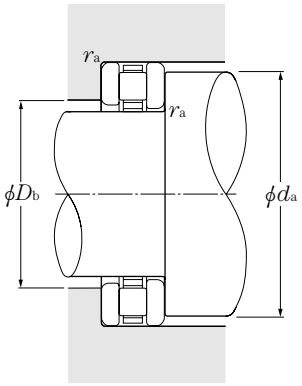
Type K893
(Thrust needle roller and cage assy)



WS bearing ring
(Inner ring)

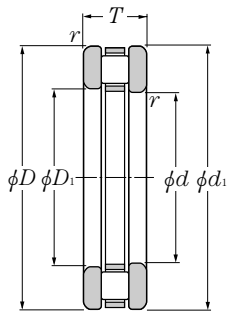


GS bearing ring
(Outer ring)

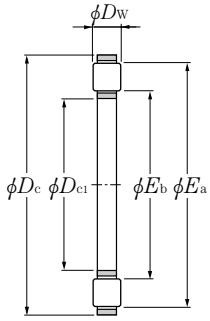


Bearing numbers				Reference dimensions mm		Abutment dimensions mm			Mass kg		
bearing	thrust cylindrical roller and cage assembly	inner ring	outer ring	E_b	E_a	d_a min	D_b max	r_{as} max	811 812 893	K811 K812 K893	WS GS
81113T2	**K81113T2**	**WS81113**	**GS81113**	68.8	85.2	85	70	1	0.340	0.090	0.125
81213T2	**K81213T2**	**WS81213**	**GS81213**	69.9	93.3	93	71	1	0.775	0.215	0.280
89313	**K89313**	**WS89313**	**GS89313**	73.9	110.3	110	74	1	1.430	0.260	0.580
81114T2	**K81114T2**	**WS81114**	**GS81114**	73.7	90.1	90	74	1	0.365	0.097	0.135
81214T2	**K81214T2**	**WS81214**	**GS81214**	75.0	98.4	98	76	1	0.815	0.225	0.295
89314	**K89314**	**WS89314**	**GS89314**	79.8	120.2	120	80	1	1.930	0.340	0.800
81115T2	**K81115T2**	**WS81115**	**GS81115**	78.7	95.1	95	80	1	0.425	0.115	0.155
81215T2	**K81215T2**	**WS81215**	**GS81215**	80.1	103.7	103	81	1	0.860	0.240	0.310
89315	**K89315**	**WS89315**	**GS89315**	84.7	129.2	129	85	1.5	2.410	0.470	0.970
81116T2	**K81116T2**	**WS81116**	**GS81116**	83.7	100.1	100	85	1	0.445	0.119	0.165
81216T2	**K81216T2**	**WS81216**	**GS81216**	84.8	108.4	106	86	1	0.950	0.250	0.350
89316	**K89316**	**WS89316**	**GS89316**	89.8	134.2	134	90	1.5	2.530	0.490	1.020
81117T2	**K81117T2**	**WS81117**	**GS81117**	88.7	105.3	105	89	1	0.475	0.125	0.175
81217	**K81217**	**WS81217**	**GS81217**	92.2	116.9	116	92	1	1.280	0.300	0.490
89317	**K89317**	**WS89317**	**GS89317**	95.8	144.2	144	96	1.5	3.140	0.590	1.280
81118T2	**K81118T2**	**WS81118**	**GS81118**	94.7	114.3	114	95	1	0.670	0.170	0.250
81218J	**K81218J**	**WS81218**	**GS81218**	97.9	126.7	126	97	1	1.820	0.540	0.640
89318	**K89318**	**WS89318**	**GS89318**	100.8	149.2	149	101	1.5	3.280	0.620	1.330
81120T2	**K81120T2**	**WS81120**	**GS81120**	105.1	128.7	128	106	1	1.000	0.300	0.350
81220	**K81220**	**WS81220**	**GS81220**	109.2	140.0	139	109	1	2.240	0.620	0.810
89320	**K89320**	**WS89320**	**GS89320**	110.6	163.0	163	110	1.5	4.190	0.810	1.690
81122T2	**K81122T2**	**WS81122**	**GS81122**	115.0	138.8	138	116	1	1.100	0.325	0.385
81222	**K81222**	**WS81222**	**GS81222**	119.2	150.0	149	119	1	2.450	0.685	0.880
89322	**K89322**	**WS89322**	**GS89322**	122.5	183.0	183	122	2	6.030	1.150	2.440
81124T2	**K81124T2**	**WS81124**	**GS81124**	125.0	148.8	148	126	1	1.170	0.340	0.415
81224	**K81224**	**WS81224**	**GS81224**	129.2	160.0	159	129	1	2.690	0.730	0.980
81126	**K81126**	**WS81126**	**GS81126**	137.7	162.4	162	137	1	1.740	0.415	0.663
81226	**K81226**	**WS81226**	**GS81226**	140.1	179.0	173	140	1.5	4.200	1.140	1.530

Type 811
Type 812



Type 811
Type 812
(Bearing)

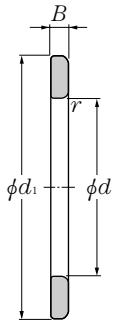


Type K811, type K812
(Thrust needle roller and cage assy)

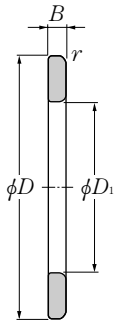
d 140～160mm

Boundary dimensions mm										Basic load ratings dynamic N	Basic load ratings static N	Basic load ratings dynamic kgf	Basic load ratings static kgf	Limiting speeds r/min	
d	D	d_1 $^{-0.2}_{-0.5}$	D_1 $^{+0.5}_{+0.2}$	T	D_{c1} E11	D_c a13	D_w $^{0}_{-0.010}$	B h11	$r_{s\,min}$ [1]	C_a	C_{oa}	C_a	C_{oa}	grease	oil
140	180	178	142	31	140	180	12	9.5	1	206 000	815 000	21 000	83 000	380	1 500
	200	197	143	46	140	200	19	13.5	1.5	370 000	1 280 000	38 000	130 000	350	1 400
150	190	188	152	31	150	190	12	9.5	1	214 000	870 000	21 800	89 000	350	1 400
160	200	198	162	31	160	200	12	9.5	1	221 000	930 000	22 600	95 000	330	1 300

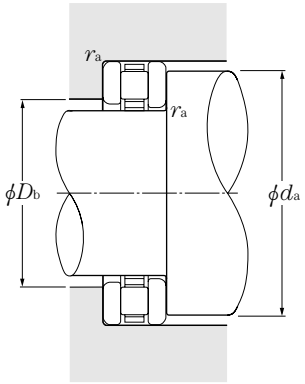
Note 1) Allowable minimum chamfer dimension r .



WS bearing ring
(Inner ring)

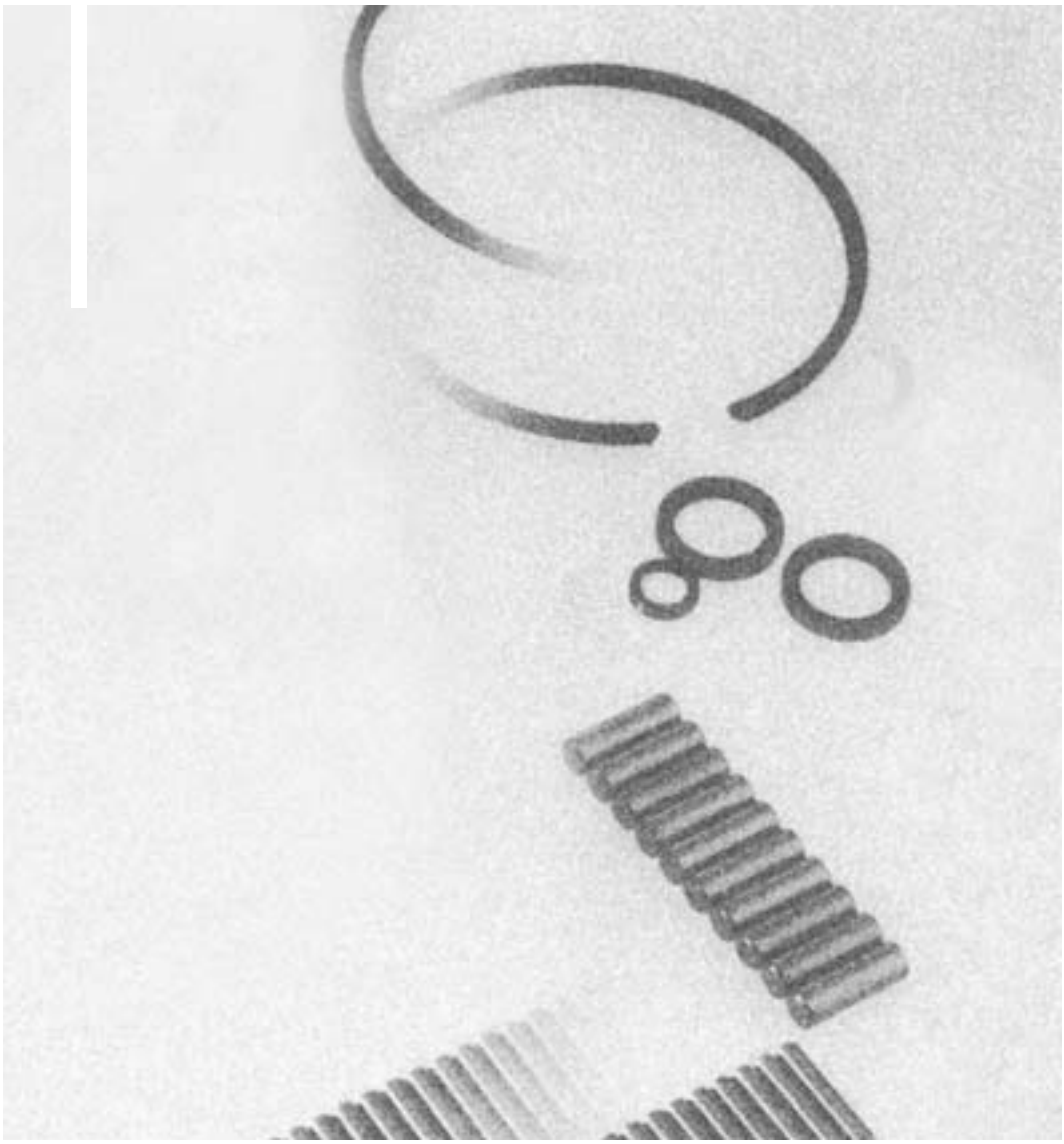


GS bearing ring
(Outer ring)



Bearing numbers				Reference dimensions mm		Abutment dimensions mm			Mass kg		
bearing	thrust cylindrical roller and cage assembly	inner ring	outer ring	E_b	E_a	d_a min	D_b max	r_{as} max	811 812 893	K811 K812 K893	WS GS
81128	**K81128**	**WS81128**	**GS81128**	147.8	172.5	172	147	1	1.950	0.450	0.750
81228	**K81228**	**WS81228**	**GS81228**	150.1	189.0	188	150	1.5	4.570	1.200	1.690
81130	**K81130**	**WS81130**	**GS81130**	157.7	182.4	182	157	1	2.070	0.470	0.800
81132	**K81132**	**WS81132**	**GS81132**	167.8	192.5	192	167	1	2.190	0.500	0.840

COMPONENTS
Needle Rollers/Snap Rings/Seals



Needle Rollers

NTN needle rollers are all made of high carbon chrome bearing steel, fine-finished by grinding and polishing after heat-treated, and the hardness thereof ranges from HRC60 to 65. These needle rollers are supplied as not only rolling element but also pin and shaft individuals.

End face profile of needle roller

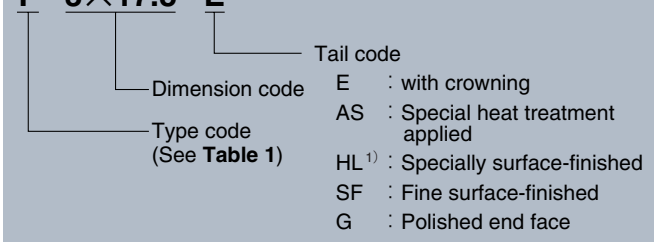
F-type needle roller bearing with flat end face is standard type, while A-type with round end face is semi-standard type. In addition to these two types, another needle roller type (nominal number with tail code E) capable of damping edge load is also available. Feel free to contact **NTN** for the detail thereof.

Table 1 End face profile

Type	Name	Profile
F	Flat	
A	Round	

Composition of nominal code

The nominal number comprises type code (end face profile), dimension code [diameter (D_w)×length (L_w)] and tail code.



1) For HL refer to Subsection 11.1 “HL Bearings” on page A-52.

Fig. 1

Needle roller tolerances

Any needle rollers are manufactured with dimensional accuracy and profile accuracy conforming to JIS B 1506 (Rollers for Roller Bearings). (**Table 2**)

Table 2 Needle roller tolerances Unit: μm

Characteristics	Tolerance and allowable value
Tolerance for mean value of diameter D_w	0～−10
Mutual deviation of diameter D_w	2
Roundness of diameter D_w, variation in diameter	1.0 ($L_w/D_w \leqq 6$) 1.5 ($L_w/D_w > 6$)
Tolerance for length L_w	h13
Accuracy class	Class-2

The needle rollers are delivered contained in same package after the mutual deviation of diameter D_w was assorted to 2 μm and less. Before being delivered, the needle rollers are identified by label colors such as red, dark blue, blue, etc. according to the respective dimensional tolerances.

Further, mixed use of needle rollers contained in packages of different label colors is prohibited.

Table 3 Discrimination of needle rollers

Label color	Dimensional tolerance range μm	Discrimination
Red	0～− 2	Standard
Dark blue	−1～− 3	
Blue	−2～− 4	
Black	−3～− 5	
White	−4～− 6	
Gray	−5～− 7	Semi-standard
Green	−6～− 8	
Brown	−7～− 9	
Yellow	−8～−10	

Application of needle rollers

When composing a full-complement roller type needle roller bearing using standard needle rollers, shaft diameter (d), housing hole diameter (D), circumferential clearance (ΔC)

and radial clearance (Δr) must be determined respectively. These can be calculated as follows from needle roller diameter (D_w) and number of rollers (Z). (Refer to **Fig. 2**)

Determine the minimum value of circumferential clearance (ΔC) using **formula (1)**. And select radial clearance (Δr) according to applicable shaft diameter and operating conditions, in reference to **Table 5.1**, as the guideline, of 5.1 "**Bearing Radial Clearance**" (page A-30). In general, full-complement roller type bearing needs radial clearance larger than that in needle roller and cage assembly type roller bearing.

$$\Delta C = (0.005 \sim 0.020) \times Z \text{ mm (minimum value)} \quad \textbf{(1)}$$

Then, determine the minimum value of housing bore diameter (D) and the maximum value of shaft diameter (d) using the **formulas (2)** and **(3)**.

$$D = \left[\frac{1}{\sin\left(\frac{\pi}{Z}\right)} + 1 \right] \cdot D_w + \frac{\Delta C}{\pi} \text{ mm (minimum value)} \quad \cdots\cdots\cdots\cdots\textbf{(2)}$$

$$d = D - 2D_w - \Delta r \text{ mm (maximum value)} \cdots\cdots\cdots\textbf{(3)}$$

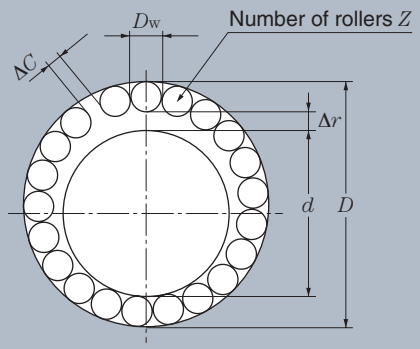


Fig. 2

The maximum value of housing bore diameter (D) required to retain needle rollers in a housing by a keystone can be determined from the minimum diameter value ($D_{w\,min}$) of the roller and the number of rollers using **formula (4)**. The coefficient K to be used in that time is as shown in **Table 4**.

$$D = K \cdot D_{w\,min} \text{ mm (max.)} \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\textbf{(4)}$$

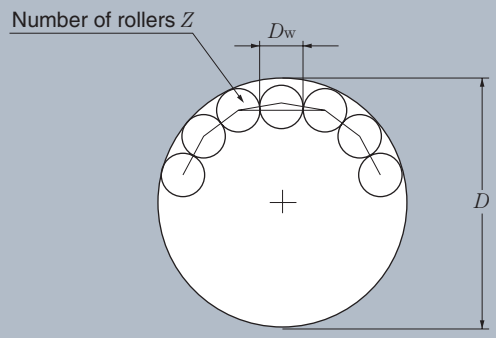
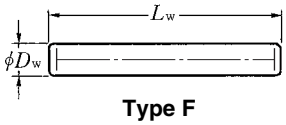


Fig. 3

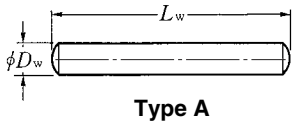
Table 4 Coefficient K value

Z	K	Z	K
8	3.6763333	17	6.4536463
9	3.9709394	18	6.7689303
10	4.2727719	19	7.0846088
11	4.5789545	20	7.4006100
12	4.8879667	21	7.7168786
13	5.1989251	22	8.0333713
14	5.5112799	23	8.3500534
15	5.8246707	24	8.6668970
16	6.1388508	25	8.9838796

Type F
Type A



Type F



Type A

D_w 1.5～4.5mm

Boundary dimensions mm D_w	L_w	Bearing numbers flat type	Bearing numbers disk type	Mass kg per 1 000 pcs
1.5	5.8	F1.5×5.8	A1.5× 5.8	0.080
	6.8	F1.5×6.8	A1.5× 6.8	0.090
	7.8	F1.5×7.8	A1.5× 7.8	0.104
	9.8	F1.5×9.8	A1.5× 9.8	0.131
	11.8	F1.5×11.8	A1.5×11.8	0.159
	13.8	F1.5×13.8	A1.5×13.8	0.186
2	6.8	F2 ×6.8	A2 × 6.8	0.158
	7.8	F2 ×7.8	A2 × 7.8	0.183
	9.8	F2 ×9.8	A2 × 9.8	0.232
	11.8	F2 ×11.8	A2 ×11.8	0.281
	13.8	F2 ×13.8	A2 ×13.8	0.330
	15.8	F2 ×15.8	A2 ×15.8	0.379
	17.8	F2 ×17.8	A2 ×17.8	0.428
	19.8	F2 ×19.8	A2 ×19.8	0.477
2.5	7.8	F2.5×7.8	A2.5× 7.8	0.284
	9.8	F2.5×9.8	A2.5× 9.8	0.351
	11.8	F2.5×11.8	A2.5×11.8	0.438
	13.8	F2.5×13.8	A2.5×13.8	0.514
	15.8	F2.5×15.8	A2.5×15.8	0.591
	17.8	F2.5×17.8	A2.5×17.8	0.668
	19.8	F2.5×19.8	A2.5×19.8	0.745
	21.8	F2.5×21.8	A2.5×21.8	0.821
	23.8	F2.5×23.8	A2.5×23.8	0.898
3	9.8	F3 ×9.8	A3 × 9.8	0.556
	11.8	F3 ×11.8	A3 ×11.8	0.671
	13.8	F3 ×13.8	A3 ×13.8	0.784
	15.8	F3 ×15.8	A3 ×15.8	0.897
	17.8	F3 ×17.8	A3 ×17.8	1.01
	19.8	F3 ×19.8	A3 ×19.8	1.12
	21.8	F3 ×21.8	A3 ×21.8	1.23
3	23.8	F3 ×23.8	A3 ×23.8	1.34
	25.8	F3 ×25.8	A3 ×25.8	1.45
	27.8	F3 ×27.8	A3 ×27.8	1.56
3.5	11.8	F3.5×11.8	A3.5×11.8	0.849
	13.8	F3.5×13.8	A3.5×13.8	1.00
	15.8	F3.5×15.8	A3.5×15.8	1.15
	17.8	F3.5×17.8	A3.5×17.8	1.30
	19.8	F3.5×19.8	A3.5×19.8	1.45
	21.8	F3.5×21.8	A3.5×21.8	1.60
	23.8	F3.5×23.8	A3.5×23.8	1.75
	25.8	F3.5×25.8	A3.5×25.8	1.90
	29.8	F3.5×29.8	A3.5×29.8	2.20
	31.8	F3.5×31.8	A3.5×31.8	2.35
	34.8	F3.5×34.8	A3.5×34.8	2.58
4	13.8	F4 ×13.8	A4 ×13.8	1.27
	15.8	F4 ×15.8	A4 ×15.8	1.50
	17.8	F4 ×17.8	A4 ×17.8	1.70
	19.8	F4 ×19.8	A4 ×19.8	1.89
	21.8	F4 ×21.8	A4 ×21.8	2.09
	23.8	F4 ×23.8	A4 ×23.8	2.26
	25.8	F4 ×25.8	A4 ×25.8	2.48
	27.8	F4 ×27.8	A4 ×27.8	2.68
	29.8	F4 ×29.8	A4 ×29.8	2.87
	31.8	F4 ×31.8	A4 ×31.8	3.07
	34.8	F4 ×34.8	A4 ×34.8	3.31
	37.8	F4 ×37.8	A4 ×37.8	3.62
	39.8	F4 ×39.8	A4 ×39.8	3.82
4.5	17.8	F4.5×17.8	A4.5×17.8	2.11
	19.8	F4.5×19.8	A4.5×19.8	2.36
	21.8	F4.5×21.8	A4.5×21.8	2.61

D_w 4.5～5mm

Boundary dimensions mm		Bearing numbers		Mass kg per 1 000 pcs
D_w	L_w	flat type	disk type	
4.5	23.8	**F4.5×23.8**	A4.5×23.8	2.86
	25.8	**F4.5×25.8**	A4.5×25.8	3.11
	29.8	**F4.5×29.8**	A4.5×29.8	3.62
	31.8	**F4.5×31.8**	A4.5×31.8	3.87
	34.8	**F4.5×34.8**	A4.5×34.8	4.25
	37.8	**F4.5×37.8**	A4.5×37.8	4.63
	39.8	**F4.5×39.8**	A4.5×39.8	4.88
	44.8	**F4.5×44.8**	A4.5×44.8	5.51
5	19.8	**F5 ×19.8**	A5 ×19.8	2.89
	21.8	**F5 ×21.8**	A5 ×21.8	3.20
	23.8	**F5 ×23.8**	A5 ×23.8	3.52
	25.8	**F5 ×25.8**	A5 ×25.8	3.82
	29.8	**F5 ×29.8**	A5 ×29.8	4.45
	31.8	**F5 ×31.8**	A5 ×31.8	4.74
	34.8	**F5 ×34.8**	A5 ×34.8	5.11
	37.8	**F5 ×37.8**	A5 ×37.8	5.55
	39.8	**F5 ×39.8**	A5 ×39.8	5.85
	49.8	**F5 ×49.8**	A5 ×49.8	7.33

Snap Rings

These snap rings are used exclusively for fixing or guiding a needle roller bearing ring or cage in axial direction. Furthermore, these snap rings have the profile identical to C-type concentric snap ring specified in JIS B 2806 and, in addition, snap rings with smaller section height (b) and also available in smaller dimension range are manufactured according to application of needle roller bearings. These snap rings are manufactured using hard steel wire rod and, after manufactured, chemical conversion treatment is applied to the surface thereof.

Types of snap ring

Two types of snap ring are available; one is **Type WR** designed for application to shaft and another is **Type BR** for application to bearing housing.

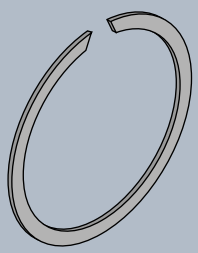


Fig. 1 WR snap ring

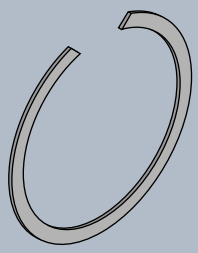


Fig. 2 BR snap ring

Table 1 Cut section angle of snap ring (α)

Type	Nominal number	Cut section angle(α)
Type WR	WR4, WR5	40°
	Type WR6 up to	60°
Type BR	All nominal numbers	90°

Composition of nominal number

The snap ring nominal number is composed of type code (**WR** or **BR**) and dimension code. The dimension code represents applicable shaft diameter in **Type WR** and applicable housing bore diameter in **Type BR.**

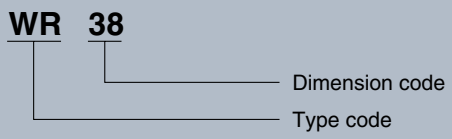


Fig. 3

Allowable running speed

The allowable running speed for **Type WR** for shaft application is as described in applicable Dimensions Table. This means the running speed when snap ring begins to get loose with opening of its cut ends.

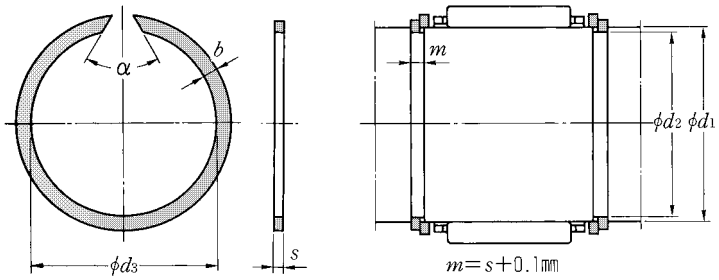
Mounting relation

It is recommended to insert a spacer between snap ring and cage for guiding the cage in axial direction. On occasion, snap ring is difficult to remove, but limited to a portion in which a pull-out

tool can not be inserted easily. In such a case, it is necessary to consider the use of general snap rings specified in JIS B 2804 (C-type snap rings) or B 2806 (C-type concentric snap rings).

For shaft

Type WR

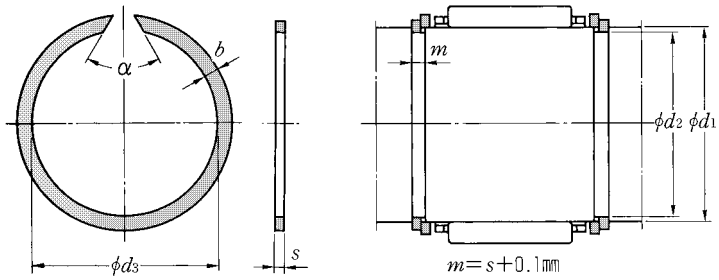


Bearing numbers	Boundary dimensions mm d_1	d_3 max	b	s ±0.06	d_2		Limiting speeds r/min
WR 4	4	3.7	0.8	0.5	3.8		275 000
WR 5	5	4.7	1	0.5	4.8		192 000
WR 6	6	5.6	1.1	0.7	5.7		141 000
WR 7	7	6.5	1.2	0.7	6.7	$^{0}_{-0.09}$	134 000
WR 8	8	7.4	1.3	1	7.6		108 000
WR 9	9	8.4	1.3	1	8.6		80 000
WR10	10	9.4	1.3	1	9.6		68 000
WR11	11	10.2	1.3	1	10.5		64 500
WR12	12	11.2	1.3	1	11.5		53 000
WR13	13	12.2	1.3	1	12.5		49 600
WR14	14	13.1	1.5	1.2	13.5	$^{0}_{-0.11}$	45 900
WR15	15	14	1.75	1.2	14.4		44 500
WR16	16	15	1.75	1.2	15.4		38 000
WR17	17	16	1.75	1.2	16.4		34 500
WR18	18	17	1.75	1.2	17.4		30 000
WR19	19	17.9	1.75	1.2	18.4		28 900
WR20	20	18.7	1.75	1.2	19.2		26 200
WR21	21	19.7	1.75	1.2	20.2		23 400
WR22	22	20.7	1.75	1.2	21.2		20 800
WR23	23	21.7	1.75	1.2	22.2		19 500
WR24	24	22.5	1.75	1.2	23	$^{0}_{-0.13}$	18 100
WR25	25	23.5	1.75	1.2	24		16 400
WR26	26	24.5	1.75	1.2	25		14 800
WR28	28	26.5	2.3	1.5	27		15 400
WR29	29	27.5	2.3	1.5	28		14 400
WR30	30	28.5	2.3	1.5	29		13 200
WR32	32	30.2	2.3	1.5	30.8	$^{0}_{-0.16}$	13 300
WR35	35	33.2	2.3	1.5	33.8		10 700

Bearing numbers	Boundary dimensions mm d_1	d_3 max	b	s ±0.06	d_2		Limiting speeds r/min
WR 37	37	35.2	2.3	1.5	35.8		9 200
WR 38	38	36.2	2.3	1.5	36.8		8 700
WR 40	40	37.8	2.3	1.5	38.5		8 100
WR 42	42	39.8	2.3	1.5	40.5		7 000
WR 43	43	40.8	2.3	1.5	41.5	$^{0}_{-0.16}$	6 800
WR 45	45	42.8	2.3	1.5	43.5		5 800
WR 47	47	44.8	2.3	1.5	45.5		5 500
WR 48	48	45.8	2.3	1.5	46.5		5 300
WR 50	50	47.8	2.3	1.5	48.5		4 800
WR 52	52	49.8	2.3	1.5	50.5		4 300
WR 55	55	52.6	2.3	1.5	53.5		4 400
WR 58	58	55.6	2.3	1.5	56.5		3 900
WR 60	60	57.6	2.3	1.5	58.5		3 500
WR 61	61	58.6	2.3	1.5	59.5		3 300
WR 62	62	59.6	2.3	1.5	60.5		3 200
WR 63	63	60.6	2.3	1.5	61.5		3 100
WR 64	64	61.6	2.3	1.5	62.5	$^{0}_{-0.19}$	2 900
WR 65	65	62.6	2.3	1.5	63.5		2 800
WR 68	68	65.4	2.8	2	66.2		2 900
WR 70	70	67.4	2.8	2	68.2		2 700
WR 72	72	69.4	2.8	2	70.2		2 600
WR 73	73	70.4	2.8	2	71.2		2 500
WR 75	75	72.4	2.8	2	73.2		2 300
WR 80	80	77.4	2.8	2	78.2		1 950
WR 85	85	82	3.4	2.5	83		2 300
WR 90	90	87	3.4	2.5	88		2 000
WR 95	95	92	3.4	2.5	93	$^{0}_{-0.22}$	1 750
WR100	100	97	3.4	2.5	98		1 560

For shaft

Type WR

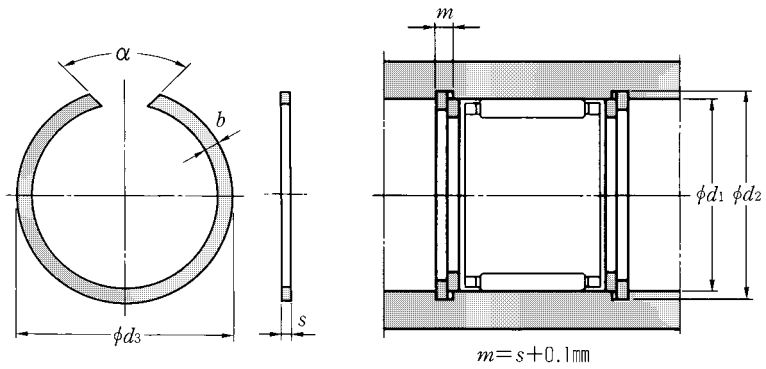


Bearing numbers	Boundary dimensions mm						Limiting speeds r/min
	d_1	d_3 max	b	s ±0.06	d_2		
WR105	105	101.7	3.4	2.5	102.7		1 390
WR110	110	106.7	3.4	2.5	107.7	$^{0}_{-0.22}$	1 240
WR115	115	111.7	3.4	2.5	112.7		1 100
WR120	120	116.7	3.4	2.5	117.7		1 000
WR125	125	121.7	3.4	2.5	122.7		920
WR130	130	126.7	3.4	2.5	127.7		830
WR135	135	131.6	4	2.5	132.4		830
WR140	140	136.6	4	2.5	137.4		760
WR145	145	141.6	4	2.5	142.4		690
WR150	150	146.6	4	2.5	147.4	$^{0}_{-0.25}$	640
WR155	155	151.6	4	2.5	152.4		590
WR160	160	156.6	4	2.5	157.4		550
WR165	165	161.6	4	2.5	162.4		500
WR170	170	166.6	4	2.5	167.4		470
WR175	175	171.6	4	2.5	172.4		440
WR180	180	175.6	5	3	177		430
WR185	185	180.6	5	3	182		590
WR190	190	185.6	5	3	187		540
WR195	195	190.6	5	3	192		510
WR200	200	195.6	5	3	197		480
WR210	210	205.6	5	3	207		420
WR220	220	215.6	5	3	217	$^{0}_{-0.29}$	380
WR225	225	220.6	5	3	222		360
WR230	230	225.6	5	3	227		350
WR240	240	235.6	5	3	237		310
WR250	250	245.6	5	3	247		270
WR260	260	253	7.5	4	255		430
WR265	265	258	7.5	4	260	$^{0}_{-0.32}$	410

Bearing numbers	Boundary dimensions mm						Limiting speeds r/min
	d_1	d_3 max	b	s ±0.06	d_2		
WR270	270	263	7.5	4	265		380
WR280	280	273	7.5	4	275		360
WR285	285	278	7.5	4	280		350
WR290	290	283	7.5	4	285	$^{0}_{-0.32}$	340
WR300	300	293	7.5	4	295		300
WR305	305	298	7.5	4	300		290
WR310	310	303	7.5	4	305		280
WR320	320	313	7.5	4	315		260
WR330	330	323	7.5	4	325		240
WR340	340	333	7.5	4	335		220
WR350	350	343	7.5	4	345		210
WR360	360	353	7.5	4	355	$^{0}_{-0.36}$	190
WR370	370	363	7.5	4	365		180
WR380	380	373	7.5	4	375		170
WR390	390	383	7.5	4	385		160
WR400	400	393	7.5	4	395		150

For housing

Type BR

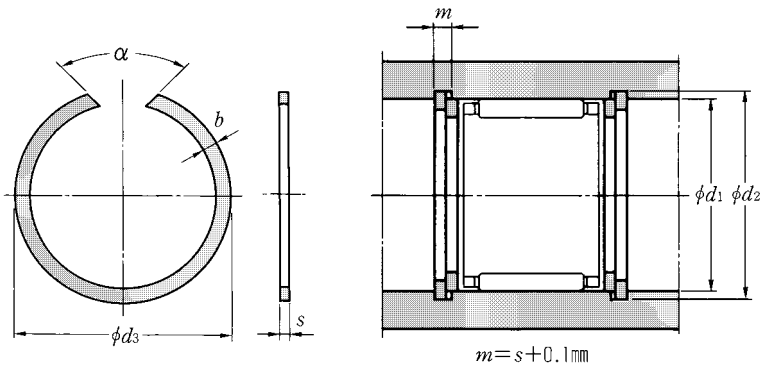


Bearing numbers	Boundary dimensions mm					
	d_1	d_3 min	b	s ±0.06	d_2	
BR 7	7	7.5	1	0.8	7.3	+0.09 0
BR 8	8	8.5	1	0.8	8.3	
BR 9	9	9.5	1.1	0.8	9.3	
BR10	10	10.6	1.2	0.8	10.4	
BR11	11	11.6	1.3	1	11.4	
BR12	12	12.7	1.3	1	12.4	
BR13	13	13.8	1.3	1	13.5	+0.11 0
BR14	14	14.8	1.3	1	14.5	
BR15	15	15.8	1.3	1	15.5	
BR16	16	16.8	1.6	1.2	16.5	
BR17	17	17.8	1.6	1.2	17.5	
BR18	18	18.9	1.75	1.2	18.5	
BR19	19	19.9	1.75	1.2	19.6	
BR20	20	21	1.75	1.2	20.6	
BR21	21	22	1.75	1.2	21.6	
BR22	22	23	1.75	1.2	22.6	
BR23	23	24	1.75	1.2	23.6	+0.13 0
BR24	24	25.2	1.75	1.2	24.8	
BR25	25	26.2	1.75	1.2	25.8	
BR26	26	27.2	1.75	1.2	26.8	
BR27	27	28.2	1.75	1.2	27.8	
BR28	28	29.2	1.75	1.2	28.8	
BR29	29	30.2	1.75	1.2	29.8	
BR30	30	31.4	2.3	1.5	31	
BR31	31	32.4	2.3	1.5	32	+0.16 0
BR32	32	33.4	2.3	1.5	33	
BR33	33	34.4	2.3	1.5	34	
BR34	34	35.4	2.3	1.5	35	

Bearing numbers	Boundary dimensions mm					
	d_1	d_3 min	b	s ±0.06	d_2	
BR35	35	36.4	2.3	1.5	36	
BR36	36	37.8	2.3	1.5	37.2	
BR37	37	38.8	2.3	1.5	38.2	
BR38	38	39.8	2.3	1.5	39.2	
BR39	39	40.8	2.3	1.5	40.2	
BR40	40	41.8	2.3	1.5	41.2	
BR41	41	42.8	2.3	1.5	42.2	+0.16 0
BR42	42	43.8	2.3	1.5	43.2	
BR43	43	44.8	2.3	1.5	44.2	
BR44	44	45.8	2.3	1.5	45.2	
BR45	45	46.8	2.3	1.5	46.2	
BR46	46	47.8	2.3	1.5	47.2	
BR47	47	48.8	2.3	1.5	48.2	
BR48	48	49.8	2.3	1.5	49.2	
BR49	49	50.8	2.3	1.5	50.2	
BR50	50	51.8	2.3	1.5	51.2	
BR52	52	54.3	2.3	1.5	53.5	
BR53	53	55.3	2.3	1.5	54.5	
BR54	54	56.3	2.3	1.5	55.5	
BR55	55	57.3	2.3	1.5	56.5	
BR57	57	59.3	2.3	1.5	58.5	+0.19 0
BR58	58	60.3	2.3	1.5	59.5	
BR60	60	62.3	2.3	1.5	61.5	
BR61	61	63.3	2.3	1.5	62.5	
BR62	62	64.3	2.3	1.5	63.5	
BR63	63	65.3	2.3	1.5	64.5	
BR64	64	66.3	2.3	1.5	65.5	
BR65	65	67.3	2.3	1.5	66.5	

For housing

Type BR



Bearing numbers	Boundary dimensions mm					
	d_1	d_3 min	b	s ±0.06	d_2	
BR 66	66	68.3	2.3	1.5	67.5	
BR 68	68	70.3	2.3	1.5	69.5	
BR 70	70	72.3	2.3	1.5	71.5	
BR 72	72	74.6	2.8	2	73.8	
BR 73	73	75.6	2.8	2	74.8	+0.19 0
BR 74	74	76.6	2.8	2	75.8	
BR 75	75	77.6	2.8	2	76.8	
BR 76	76	78.6	2.8	2	77.8	
BR 77	77	79.6	2.8	2	78.8	
BR 78	78	80.6	2.8	2	79.8	
BR 79	79	81.6	2.8	2	80.8	
BR 81	81	83.6	2.8	2	82.8	
BR 82	82	84.6	2.8	2	83.8	
BR 83	83	85.6	2.8	2	84.8	
BR 85	85	87.6	2.8	2	86.8	
BR 86	86	88.6	2.8	2	87.8	
BR 88	88	91	3.4	2.5	90	
BR 90	90	93	3.4	2.5	92	
BR 92	92	95	3.4	2.5	94	+0.22 0
BR 93	93	96	3.4	2.5	95	
BR 95	95	98	3.4	2.5	97	
BR 97	97	100	3.4	2.5	99	
BR 98	98	101	3.4	2.5	100	
BR100	100	103	3.4	2.5	102	
BR102	102	105.3	3.4	2.5	104.3	
BR103	103	106.3	3.4	2.5	105.3	
BR105	105	108.3	3.4	2.5	107.3	
BR107	107	110.3	3.4	2.5	109.3	

Bearing numbers	Boundary dimensions mm					
	d_1	d_3 min	b	s ±0.06	d_2	
BR108	108	111.3	3.4	2.5	110.3	
BR110	110	113.3	3.4	2.5	112.3	
BR112	112	115.3	3.4	2.5	114.3	+0.22 0
BR113	113	116.3	3.4	2.5	115.3	
BR115	115	118.3	3.4	2.5	117.3	
BR117	117	120.3	3.4	2.5	119.3	
BR118	118	121.3	3.4	2.5	120.3	
BR120	120	123.3	3.4	2.5	122.3	
BR123	123	126.3	3.4	2.5	125.3	
BR125	125	128.3	3.4	2.5	127.3	
BR127	127	130.3	3.4	2.5	129.3	
BR130	130	133.3	3.4	2.5	132.3	
BR133	133	136.3	3.4	2.5	135.3	
BR135	135	138.3	3.4	2.5	137.3	
BR137	137	140.3	3.4	2.5	139.3	
BR140	140	143.6	4	2.5	142.6	+0.25 0
BR143	143	146.6	4	2.5	145.6	
BR150	150	153.6	4	2.5	152.6	
BR153	153	156.6	4	2.5	155.6	
BR160	160	163.6	4	2.5	162.6	
BR163	163	166.6	4	2.5	165.6	
BR165	165	168.6	4	2.5	167.6	
BR170	170	173.6	4	2.5	172.6	
BR173	173	176.6	4	2.5	175.6	
BR175	175	178.6	4	2.5	177.6	
BR180	180	183.6	4	2.5	182.6	
BR183	183	186.6	4	2.5	185.6	+0.29 0
BR190	190	194.5	5	3	193	

Bearing numbers	Boundary dimensions mm					
	d_1	d_3 min	b	s ±0.06	d_2	
BR195	195	199.5	5	3	198	
BR200	200	204.5	5	3	203	
BR205	205	209.5	5	3	208	
BR210	210	214.5	5	3	213	$^{+0.29}_{0}$
BR215	215	219.5	5	3	218	
BR220	220	224.5	5	3	223	
BR225	225	229.5	5	3	228	
BR230	230	234.5	5	3	233	
BR240	240	244.5	5	3	243	
BR250	250	254.5	5	3	253	
BR260	260	267	7.5	4	265	$^{+0.32}_{0}$
BR270	270	277	7.5	4	275	
BR280	280	287	7.5	4	285	
BR300	300	307	7.5	4	305	
BR320	320	327	7.5	4	325	
BR325	325	332	7.5	4	330	
BR350	350	357	7.5	4	355	
BR355	355	362	7.5	4	360	
BR360	360	367	7.5	4	365	$^{+0.36}_{0}$
BR375	375	382	7.5	4	380	
BR380	380	387	7.5	4	385	
BR385	385	392	7.5	4	390	
BR395	395	402	7.5	4	400	
BR400	400	407	7.5	4	405	
BR415	415	422	7.5	4	420	
BR420	420	427	7.5	4	425	$^{+0.40}_{0}$
BR440	440	447	7.5	4	445	

Seals

Seals

These are the special-purposed seals for needle roller bearings whose cross sectional height is designed so small as to match applicable needle roller bearings. These contact seals are made of synthetic rubber reinforced with steel plate, being then used in operating temperature range of -25 to +120˚C and, under continuous running, at 100˚C and less. Further, feel free to contact **NTN** for the use of these seals under special operating condition, e.g. operating temperature of over 120˚C.

Types of seal

Two different seal types are available; one is **Type G** with one lip and another is **Type GD** with two lips. In addition to these two, sliding rubber seals (**LEG**, **LEGD**), wherein lubrication property was assigned to a rubber material fulfilling low torque under an non-lubrication environment, are also manufactured. Feel free to contact **NTN** for the detail of these seal types.

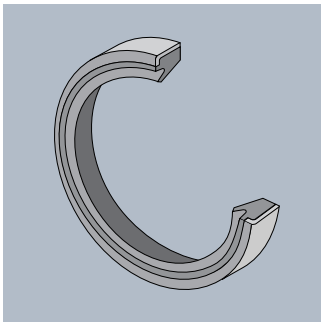


Fig. 1 Type G (LEG)

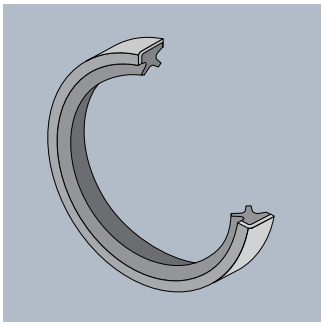


Fig. 2 Type GD (LEGD)

Where seal contact surface and lubricating condition are good, **Type G** with one lip can be used up to 10m/s maximum at peripheral speed and **Type GD** with two lips used up to 6m/s maximum.

This seal is intended to hold grease and to prevent dust invasion from outside. It has no function to hold lubrication oil, etc. unlike oil seal.

Composition of nominal number

Nominal seal number is composed of type code (G, GD) and dimension code (bore dia.×outer dia.×width).

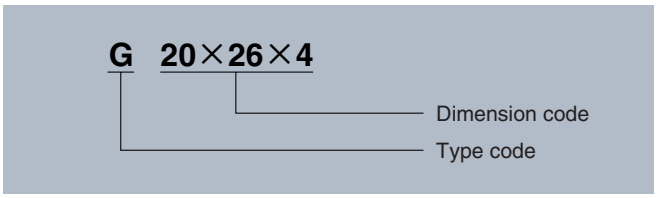


Fig. 3

Fitting relations

The seal contact surface must be finished by grinding, after hardened, to get its good sealing performance. In addition, harmful flaw and indent are not allowed to exist on the seal contact surface. Where the inner ring raceway surface is used as seal contact surface, a wide inner ring must be used.

When fitting a seal in a bearing, it is desirable to taper the shaft end face or the housing end face so as to protect the seal lip and outer surface from damaging, as illustrated in **Fig. 4**. Where shaft end is not tapered or chamfered (rounded), it is recommended to a fitting jig as illustrated in **Fig.5**. Regarding the seal to housing interference, a adequate interference can be got in the usual housing tolerance range of G7 toR7.

Furthermore, it is recommended to apply pre-coat of a lubricant to the seal lip before fitting seals G and GD, for better lubrication. Further, when fitting a seal and inserting it through a shaft, take good care to protect its lip from deforming.

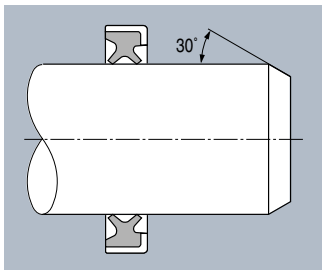


Fig. 4

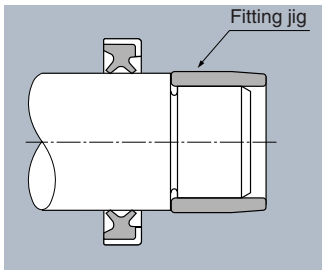


Fig. 5

Seal application examples

Figs.6 to **8** illustrate design examples using these seals.

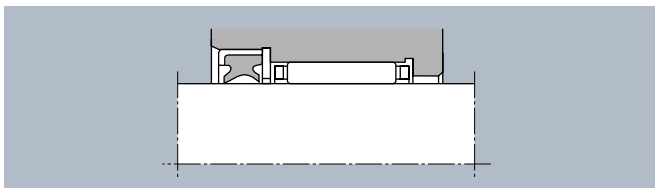


Fig. 6

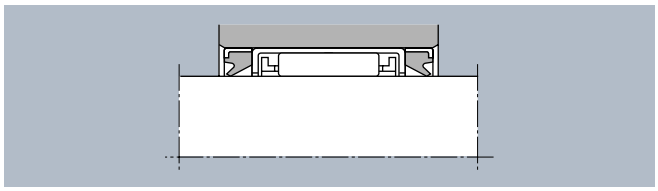


Fig. 7

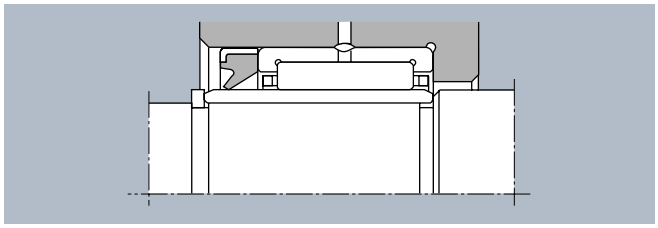
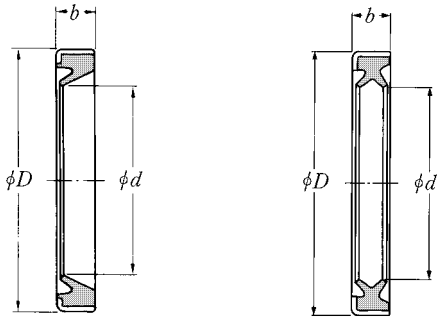


Fig. 8

Type G
Type GD

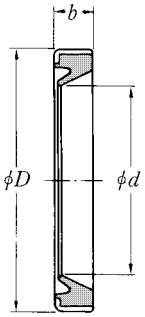


Type G　　Type GD

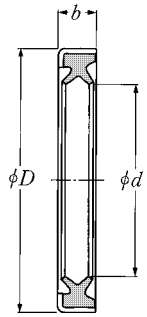
Bearing numbers		Boundary dimensions mm			Mass ×10⁻³ kg	
Type G	Type GD	d	$D^{1)}$	b $^{+0.2}_{0}$	G	GD
G 4×8×2K	— —	4	8	2	0.18	—
G 5×9×2K	— —	5	9 $^{+0.081}_{+0.023}$	2	0.19	—
G 5×10×2K	— —	5	10	2	0.22	—
G 6×10×2K	— —	6	10	2	0.21	—
G 6×12×2K	— —	6	12	2	0.38	—
G 7×11×2K	— —	7	11	2	0.25	—
G 7×14×2K	— —	7	14	2	0.52	—
G 8×12×3K	— —	8	12	3	0.41	—
G 8×15×3K	— —	8	15	3	0.74	—
G 9×13×3K	— —	9	13 $^{+0.098}_{+0.028}$	3	0.44	—
G 9×16×3K	— —	9	16	3	0.69	—
G10×14×3K	— —	10	14	3	0.50	—
G10×17×3K	— —	10	17	3	0.87	—
G12×16×3K	— —	12	16	3	0.56	—
G12×18×3K	— —	12	18	3	0.86	—
G12×19×3K	— —	12	19	3	0.94	—
G13×19×3K	— —	13	19	3	0.87	—
G14×20×3K	GD14×20×3K	14	20	3	0.96	0.99
G14×21×3K	GD14×21×3K	14	21	3	1.1	1.1
G14×22×3K	GD14×22×3K	14	22	3	1.3	1.2
G15×21×3K	GD15×21×3K	15	21	3	1.0	1.0
G15×23×3K	GD15×23×3K	15	23 $^{+0.119}_{+0.035}$	3	1.3	1.3
G16×22×3K	GD16×22×3K	16	22	3	1.3	1.1
G16×24×3K	GD16×24×3K	16	24	3	1.3	1.3
G16×25×3K	GD16×25×3K	16	25	3	1.6	1.6
G17×23×3K	GD17×23×3K	17	23	3	1.3	1.1
G17×25×3K	GD17×25×3K	17	25	3	1.5	1.4
G18×24×3K	GD18×24×3K	18	24	3	1.2	1.2
G18×26×4K	GD18×26×4K	18	26	4	1.8	1.8
G19×27×4K	GD19×27×4K	19	27	4	2.0	1.9
G20×26×4K	GD20×26×4K	20	26	4	1.8	1.8
G20×28×4K	GD20×28×4K	20	28 $^{+0.119}_{+0.035}$	4	2.1	2.1
G21×29×4K	GD21×29×4K	21	29	4	2.2	2.1
G22×28×4K	GD22×28×4K	22	28	4	1.8	1.9
G22×30×4K	GD22×30×4K	22	30	4	2.2	2.3
G24×32×4K	GD24×32×4K	24	32	4	2.5	2.4
G25×32×4K	GD25×32×4K	25	32	4	2.3	2.2
G25×33×4K	GD25×33×4K	25	33	4	2.5	2.5
G25×35×4K	GD25×35×4K	25	35	4	2.6	2.6
G26×34×4K	GD26×34×4K	26	34	4	2.6	2.6
G28×35×4K	GD28×35×4K	28	35	4	2.4	2.5
G28×37×4K	GD28×37×4K	28	37	4	3.1	2.8
G29×37×4K	GD29×37×4K	29	37	4	2.7	2.7
G29×38×4K	GD29×38×4K	29	38	4	3.2	2.9
G30×37×4K	GD30×37×4K	30	37 $^{+0.143}_{+0.043}$	4	2.7	2.6
G30×40×4K	GD30×40×4K	30	40	4	3.6	3.3
G32×42×4K	GD32×42×4K	32	42	4	3.7	3.9
G32×45×4K	GD32×45×4K	32	45	4	5.1	5.2
G35×42×4K	GD35×42×4K	35	42	4	3.0	2.9
G35×45×4K	GD35×45×4K	35	45	4	4.1	3.6
G37×47×4K	GD37×47×4K	37	47	4	4.0	3.8
G38×48×4K	GD38×48×4K	38	48	4	4.4	4.0
G40×47×4K	GD40×47×4K	40	47	4	3.3	3.5
G40×50×4K	GD40×50×4K	40	50	4	4.6	4.0
G40×52×5K	GD40×52×5K	40	52 $^{+0.173}_{+0.053}$	5	4.8	4.7
G42×52×4K	GD42×52×4K	42	52	4	4.7	4.2

Note 1) The outer diameter tolerance is the mean value of the measured values at two measuring points.

Type G
Type GD



Type G

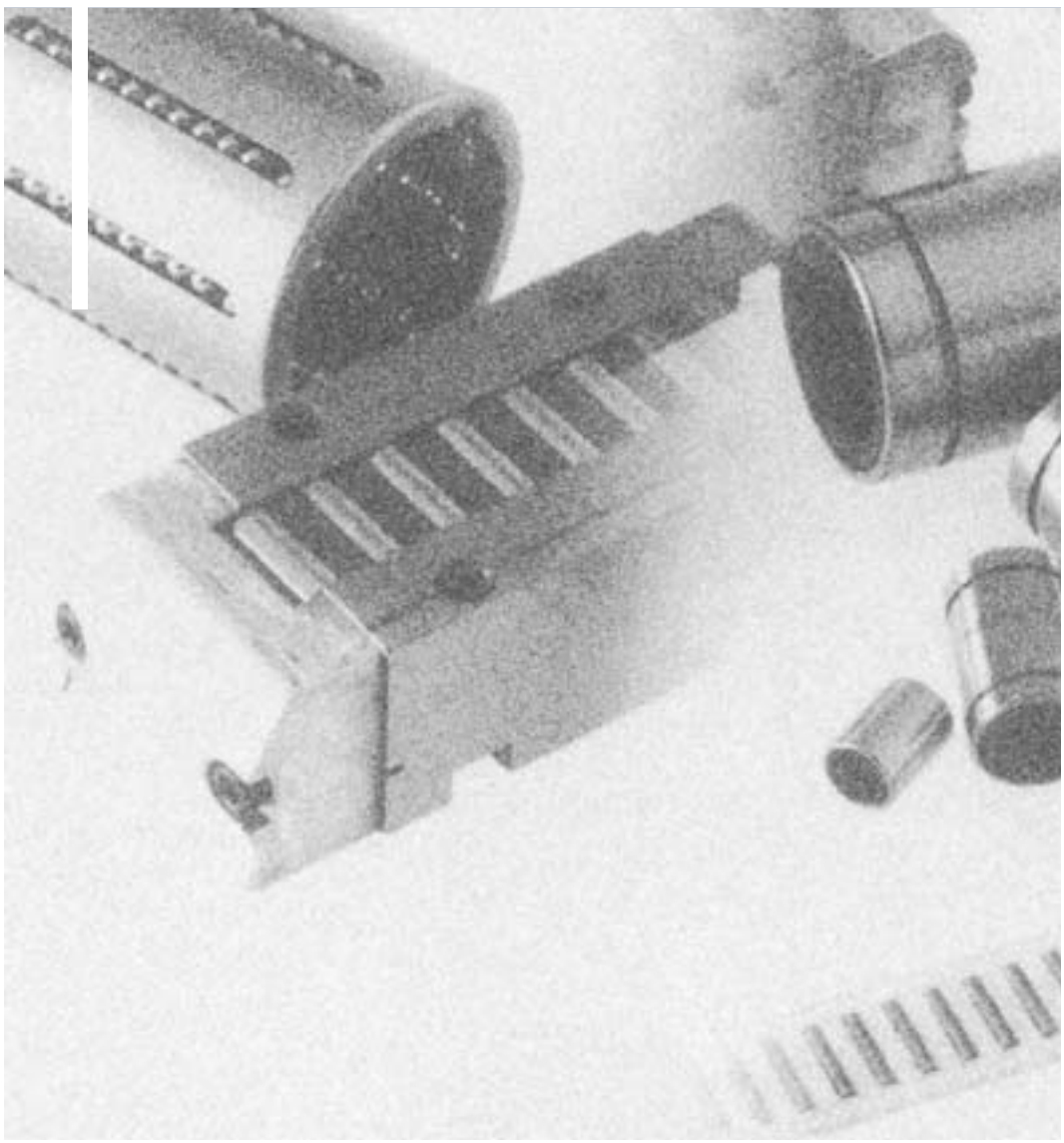


Type GD

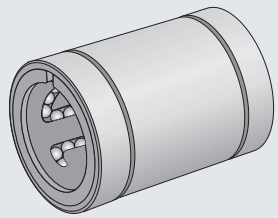
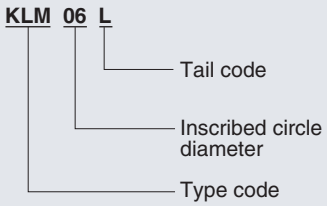
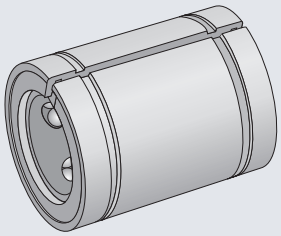
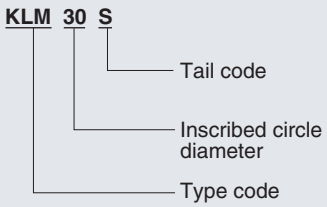
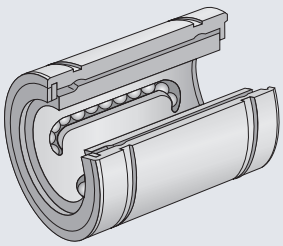
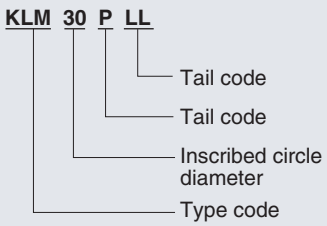
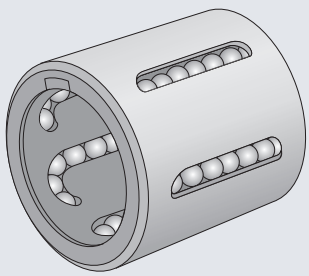
Bearing numbers		Boundary dimensions mm			Mass ×10⁻³ kg	
Type G	Type GD	d	D[1]	b $^{+0.2}_{0}$	G	GD
G43×53×4K	GD43×53×4K	43	53	4	4.8	4.3
G45×52×4K	GD45×52×4K	45	52	4	3.8	3.8
G45×55×4K	GD45×55×4K	45	55 $^{+0.173}_{+0.053}$	4	5.2	5.5
G50×58×4K	GD50×58×4K	50	58	4	4.5	5.2
G50×62×5K	GD50×62×5K	50	62	5	10.4	10

Note 1) The outer diameter tolerance is the mean value of the measured values at two measuring points.

LINEAR BALL BEARINGS:
Solid, Drawn Cup, Stroke, Linear Flat Roller, and Linear Roller Bearing Types



Linear Ball Bearings

Type		Applied shaft diameter (mm)	Composition of nominal number
KLM 	This type composed of an outer ring, steel balls and a cage is a cylindrical bearing for the most universal applications, which ensures precise and smooth infinite linear motion due to its outer ring of high rigidity.	ϕ3～ϕ40	
KLM··S 	his type is composed of an outer ring, steel balls and a cage. And both of the outer ring and the cage have an axial slit, as illustrated, so as to enable to shrink the inscribed circle diameter of the cage by pressing the outer ring in radial direction from the housing and to thereby adjust radial clearance from shaft. Thus, this type also ensures precise and smooth infinite linear motion.	ϕ16～ϕ40	
KLM··P 	This type is composed of an outer ring, steel balls and a cage. And the outer ring and the cage are of arc sectional shape, from which one row of balls (equivalent to 50˚ to 60˚ degree spacing) is removed. Thus, the arc cross-sectional ring and cage with 50˚ to 60˚ degree opening allows the bearing assy to pass through a shaft support truss or a shaft support stand on midway of the shaft stroke. This type also ensures precise and smooth infinite linear motion, similarly to other types. The bearing radial clearance can be also adjusted.	ϕ16～ϕ40	
KH 	This type is composed of an outer ring, steel balls and a cage and the outer ring is cylindrical similarly to that of Type KLM and drawn from a steel plate by precision deep drawing, then enabling to design a compact bearing construction of low section and lightweight. This type also ensures precise and smooth infinite linear motion similarly to other types.	ϕ6～ϕ50 With seal ϕ10～ϕ50	